

Stochastic Modeling of Power Purchase Agreements: A Monte Carlo-Based Techno-Economic Framework for Equity Risk in Emerging Renewable Energy Sectors

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Abstract

This paper applies a Monte Carlo simulation framework to evaluate equity investment risk in utility-scale renewable energy projects operating under Power Purchase Agreements (PPAs) in Bangladesh. The study makes two contributions. First, it develops a techno-economic analytical framework that integrates physical, financial, and contractual dimensions of project performance within a unified model, representing the first such application in the Bangladesh power sector. Second, given the limited availability of empirical data common to developing power markets, model parameters are calibrated through expert heuristics and industry benchmarks as informed proxies for the underlying data-generating processes governing revenue variability, cost dynamics, and offtaker payment behavior, enabling probabilistic inference without dependence on historical records.

Stochastic variables are assigned theoretically motivated probability distributions including normal, lognormal, and beta specifications. Two scenarios are constructed and compared: a baseline PPA structure exposed to counterparty payment risk, and an escrow-backed structure incorporating a payment security mechanism. Stochastic variables including capacity factor, tariff escalation, operation and maintenance cost shocks, and payment loss fractions are modeled using appropriate probability distributions. Simulation results, summarized through Value-at-Risk and Expected Shortfall metrics, demonstrate that escrow arrangements meaningfully reduce tail-risk exposure for equity investors. The findings carry direct implications for project structuring, regulatory design, and risk-informed capital allocation in emerging power markets.

Keywords: Power Purchase Agreement, Monte Carlo Simulation, Equity Risk Analysis, Renewable Energy Investment, Value-at-Risk, Expected Shortfall, Payment Security Mechanism, Project Finance, Emerging Markets, Bangladesh Power Sector